Notes

Posterior mean genetic liability from family history

Model, efficient inference, and evidence in `ltpred`

Bjarni Jóhann Vilhjálmsson

Aarhus University and SMARTbiomed

`https://github.com/bvilhjal/ltpred` · v0.7.5

15 October 2026

Benchmark tables retain the historical v0.7.4 evidence snapshot.

Abstract

The LT-FH family of methods estimates $\mu_i = \mathbb{E}[a_i \mid \text{family data}]$ under a Gaussian liability-threshold model. For rectangular observations that is ordinary BLUP after the liabilities have been replaced by their truncated-normal means. The same core has three uses: (I) risk prediction from family history, with own status held out and an optional downstream PGS; (II) a quantitative GWAS phenotype, with own status in; (III) genetic architecture, between-disease covariance, and the age pattern of incidence, for which h^2, r_g and $K(\cdot)$ can stand alone. LT-FH, LT-FH++ and ADuLT choose different rectangles; PA-FGRS also permits a censoring mixture of truncated laws. These are observation models, not different genetic models. These notes fix the estimand, the observation-model menu, the exactness boundary of the Pearson–Aitken (PA) approximation, and the sampling conditions under which a pedigree Haseman–Elston fit recovers h^2. They also derive exact observation reduction, at-most-two-dimensional parental-factor quadrature, selected relationship recursion, and an optional pairwise likelihood for single- and multi-trait common-threshold covariance fitting (Section 5). The following simulation summaries describe the archived benchmark snapshots; they do not validate the newly introduced fitting criterion or its uncertainty. Simulation under the model's own generative assumptions gives $\mathrm{Corr}(\mathrm{PA}, \mathrm{Gibbs}) \geq 0.9995$ without the censoring mixture, a classic-LT-FH causal-SNP noncentrality ratio of $1.47 \pm 0.04\times$ over case/control (independent SNPs), and a paired $+0.145 \pm 0.015$ LT-FH++ increment over matched ADuLT when onset is a deterministic function of liability and the CIP is known. The same personalised thresholds that supply that increment also remove a simulated cohort-confounding λ_{GC} of 16.46 ± 0.40 (Table 10). A PGS and μ_i remain complementary: joint test $R^2 = 0.338$ against 0.236 and 0.170 (Table 11). The association and personalisation figures describe optimistic, model-specific scenarios, not proved upper bounds for register data: a pin-versus-interval ablation keeps almost all of the onset increment except at high prevalence, and pinning is calibrated only when onset is the CIP inverse of liability. They are not estimates of clinical utility. On the locked comparison to the two public R packages (Section 8.2), ltpred is much more computationally efficient at the same algorithm: LTFHPlus Gibbs takes $6.79 \pm 0.08\times$ the ltpred Gibbs wall-clock, and LTFGRS PA takes $1418 \pm 30\times$ the ltpred PA wall-clock.

The intended reader already uses GWAS, liability-scale h^2, and Falconer's threshold construction. The notes do not re-derive those. They are about what is being conditioned on, where the sequential approximation is exact, which of three uses that quantity serves, and which simulation claims survive their generative assumptions.

1 Estimand

Write a_i for the additive genetic liability of a designated outbred proband, $\mathrm{Var}(a_i) = h^2$ on the usual unit-liability scale, and ℓ_F for the vector of full liabilities on the people whose disease records are used (the proband may or may not be among them). Let D_F denote the complete observation model encoded by those records. The general target is

$$\mu_i \;=\; \mathbb{E}[a_i \,|\, D_F,\; A,\; h^2,\; K(\,\cdot\,)], \tag{1}$$

where A is the additive relationship matrix and $K(\,\cdot\,)$ is the prevalence or cumulative-incidence model that turns status and age into interval endpoints or mixture weights. For LT-FH, LT-FH++ and ADuLT, D_F is the rectangular event $\ell_F \in C_F$, one interval per person. The PA-FGRS censoring mixture is a mixture of truncated laws and is not, in general, one rectangle. For an inbred target the kinship API reports the standardized genetic coordinate in Section 2; its prior variance need not be h^2.

If ℓ_F were observed continuously, (1) would be the selection index / animal-model BLUP

$$\mathbb{E}[a_i \mid \ell_F] \;=\; \mathrm{Cov}(a_i, \ell_F)\,\mathrm{Var}(\ell_F)^{-1}\,\ell_F \tag{2}$$

[6, 7]. A single relative with relationship r contributes $r\,h^2\,\ell_r$. Disease records do not give ℓ_F. They give $\ell_F \in C_F$. Under joint normality,

$$\mathbb{E}[a_i \mid \ell_F \in C_F] \;=\; \mathrm{Cov}(a_i, \ell_F)\,\mathrm{Var}(\ell_F)^{-1}\,\mathbb{E}[\ell_F \mid \ell_F \in C_F]. \tag{3}$$

The BLUP weights are unchanged; only the right-hand side is replaced by a truncated-normal mean. The map from the binary (or censored) records to μ_i is therefore nonlinear, and reduces to (2) only when every interval collapses to a point.

Two immediate consequences, both easy to get wrong in an analysis.

1. **The proband's own diagnosis is optional.** Include it when μ_i is a GWAS phenotype *constructed from* that diagnosis [9, 20]. Leave the corresponding interval as $(-\infty, \infty)$ when the same diagnosis is the outcome you will later predict or classify. Conditioning on the answer is leakage, not sophistication.

2. (1) **is not a SNP polygenic score.** No marker effects enter. Combining μ_i with a PGS is a downstream model [10, 15].

Those two remarks are the split between two uses of μ_i. A third use never needs μ_i (Table 1).

I. **Risk prediction from family history.** Own status out of D_F. An optional PGS is combined afterwards [10, 15], not inside (1).

II. **A quantitative GWAS phenotype.** Own status in. This is the LT-FH association use [9]. ADuLT is the no-relative special case [21].

III. **Architecture, relationships, aetiology.** Liability-scale h^2 and r_g, and $K(\,\cdot\,)$, are already objects of interest: genetic architecture, between-disease covariance, and the age/sex/cohort pattern of incidence. Pedigree scoring is optional. Fitting h^2 from the *same* families is a different contract (Section 7).

The practical order is the vignette's six-stage workflow: (0) supply liability-scale population parameters, (1) identify the pedigree, (2) specify population prevalence or CIPs, (3) encode the permitted observations, (4) estimate the score, and (5) perform the downstream analysis. At the low level, the caller builds the own-status inclusion or exclusion into the bounds. The

register driver performs that construction through an explicit use argument (Section 6). Multi-trait scoring accepts supplied covariance parameters. Joint fitting of heritabilities, genetic and residual environmental correlations is available under the separate contract in Section 5.5.

The multivariate threshold model on a pedigree is Sham's framework [23]. So, Kwan, Cherny and Sham [24] already combined family history with known loci, age, and competing mortality; they targeted absolute risk. Here uses I and II target the breeding value; use III may stop at h^2, r_g and $K(\cdot)$.

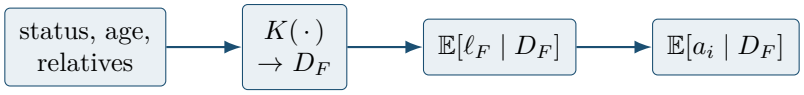


Figure 1: How D_F becomes μ_i (uses I and II). The published methods differ in how they build D_F. Rectangular methods share (3); PA-FGRS may instead use the mixture in (12). Use III can stop at h^2, r_g and $K(\cdot)$ without this pipeline.

Table 1: Three uses of the same liability-threshold core. Uses I and II need μ_i; use III can stop at h^2, r_g and/or $K(\cdot)$.

	Question	Own status / what is required
I	Risk prediction (± a PGS)	out of D_F; PGS is downstream
II	GWAS phenotype	in; ADuLT skips relatives
III	Architecture, relationships, aetiology	h^2, r_g, $K(\cdot)$; scoring optional

2 Covariance

For a non-inbred person without shared environment, the liability split is

$$\ell = a + e, \qquad a \sim \mathcal{N}(0, h^2), \quad e \sim \mathcal{N}(0, 1 - h^2), \quad a \perp\!\!\!\perp e, \tag{4}$$

so $\ell \sim \mathcal{N}(0,1)$ and a case is $\ell > T$ with $T = \Phi^{-1}(1-K)$ at a single prevalence K [25, 5]. In the same outbred, additive-only model, on a pedigree,

$$\mathrm{Cov}(a_i, a_j) = h^2 A_{ij}, \qquad \mathrm{Cov}(\ell_i, \ell_j) = h^2 A_{ij} \ (i \neq j), \qquad \mathrm{Var}(\ell_i) = 1, \tag{5}$$

with $A_{ij} = 2\phi_{ij}$. The default role model takes mates as unrelated ($A_{mf} = 0$); it is not a generative model of assortment. Actual parent pointers may instead imply related mates and inbreeding, handled by the tabular method [8].

With optional shared environment, form the full raw covariance

$$V = h^2 A + c^2 C + m^2 M + e^2 I, \qquad e^2 = 1 - h^2 - c^2 - m^2 \geq 0. \tag{6}$$

The environmental kernels C and M have unit diagonals. Hence $V_{ii} = 1$ for an outbred person, while inbreeding gives $V_{ii} = 1 + h^2(A_{ii} - 1)$. The kinship API keeps prevalence thresholds on a unit-variance scale by standardizing both the observed liabilities and the target:

$$\begin{aligned} s_i &= \sqrt{V_{ii}}, \qquad \Sigma_{ij} = V_{ij}/(s_i s_j), \qquad g_i = a_i/s_i, \\ \mathrm{Var}(g_i) &= h^2 A_{ii}/V_{ii}, \qquad \mathrm{Cov}(g_i, \ell_j/s_j) = h^2 A_{ij}/(s_i s_j). \end{aligned} \tag{7}$$

It reports moments of g_i, not raw a_i, when $s_i \neq 1$. Recover raw additive moments by multiplying the mean by s_i and the variance by V_{ii}, if that is the intended comparison. The target couples to relatives only through $h^2 A$, not through C or M.

Role-based scoring constructs the sibship and couple kernels from role labels. Arbitrary-kinship scoring accepts the same proportions only with caller-supplied, aligned kernels: finite, symmetric, positive semidefinite, and unit-diagonal. Additive relatedness alone cannot identify those environmental classes. The public register driver is additive-only; C/M scoring uses the lower-level kinship API.

Identification is the usual contrast arithmetic: C (full-sib environment) is the excess of sib covariance over parent–offspring (both have $A = \frac{1}{2}$); M (couple) is the covariance of genetically unrelated mates, and therefore confounds household sharing with assortment [22]. A directional maternal effect is not a symmetric kernel and is not M. An environment that scales with A is absorbed into h^2 and is not estimable. Nuclear families give roughly three contrasts; a bank of relationship-specific environments needs the relative types that extended pedigrees supply. This is the pedigree analogue of why ACE is identified from MZ/DZ twins and $A + C + D$ is not.

One inherited LTFHPlus convention is worth knowing before comparing constructors: two same-side half-sibs in the role grammar are related $\frac{1}{2}h^2$ *to each other* (implied shared second parent), and therefore also share the C environment with each other. A pedigree in which they have distinct other parents must be built from A, not from those role labels.

3 The observation-model menu

LT-FH, LT-FH++ and ADuLT are names for rectangular D_F, not for an inference algorithm. Gibbs and PA both apply to the first three. PA-FGRS, in the published specification and in the implementation, is PA-specific.

Classic LT-FH [9]. One K, one T. Cases $[T, \infty)$, controls $(-\infty, T]$, relatives included. This is the configuration-specific successor of GWAX [17].

LT-FH++ and ADuLT [20, 21]. Age, sex and cohort enter only through a CIP $K(t; s, b)$ and

$$T_i \;=\; \Phi^{-1}(1 - K(t_i; s_i, b_i)). \tag{8}$$

A control at current age c_i is truncated above at $T_i(c_i)$. A case with onset a_i is *pinned* at $T_i(a_i)$ (lower = upper). The same T_i with relatives is LT-FH++; without them, ADuLT.

Pinning is the strong assumption in this construction, and it is easy to understate. It says that the CIP is an error-free map from liability to onset, that onset dates are exact, and that two cases with the same stratum and onset have identical liability—zero residual variance given a_i. The inverse map is then the identity: the observation *is* the liability. When onset is recorded in a window, an exact likelihood must integrate over the recording bin under an explicit onset model. An interval $[T_i(a_i), \infty)$ or the lifetime interval $[T_{\text{pop}}, \infty)$ is a conservative partial-information encoding, not that general likelihood. On the same simulated families, the interval keeps almost all of the onset increment except at $K = 0.30$, where the pin adds another 0.01–0.02 in the squared-correlation proxy. When onset only tends to track liability ($\rho = 0.6$), the pin over-conditions (slope 0.92 under heavy censoring).

Because the BLUP weights in (3) do not depend on the T_i, a shared threshold error is mostly a calibration shift. A stratum-specific error also reorders scores.

Base PA-FGRS [14]. Observed cases use the *lifetime* interval $[\Phi^{-1}(1 - K_{\text{pop}}), \infty)$. Age enters for censored controls through the mixture in Section 4.3, not through the case threshold. Helpers that put cases above $T_i(a_i)$ are a different observation model; they are not PA-FGRS$_{\text{ADT}}$. Kendler's FGRS [12] is a third thing and is not implemented.

PGS and μ_i. Split $a = s + u$ with $\mathrm{Var}(s) = p := h^2_{\mathrm{SNP}}/h^2$. If two standardised scores satisfy $\mathrm{PGS} \perp\!\!\!\perp \mu \mid a$, then

$$\mathrm{Corr}(\mathrm{PGS}, \mu) \;=\; a_{\mathrm{PGS}}\, b_\mu \sqrt{p} \;=\; \sqrt{R^2_{\mathrm{PGS}}\, R^2_\mu\, p}. \tag{9}$$

The scores can each be genuinely predictive of a and still almost uncorrelated, because each is mostly noise relative to a [10, 15]. Non-overlapping discovery and family samples remove one shared noise term. They do not imply the residual is zero (ancestry, assortment, indirect effects, selection).

4 Computing the truncated mean

There is no closed form for $\mathbb{E}[\ell_F \mid C_F]$ once several intervals are live. PA and Gibbs support the general family covariance. The nuclear-family quadrature in Section 5.2 exploits additional structure.

4.1 Algorithm G (Gibbs)

Let $Q = \Sigma^{-1}$, $\mathrm{sd}_j = \sqrt{1/Q_{jj}}$, $P_{ij} = -Q_{ij}/Q_{jj}$ for $i \neq j$, and $P_{jj} = 0$. The jth conditional is then the normal $\mathcal{N}(P_{\cdot j}^\top x, \mathrm{sd}_j^2)$ restricted to (a_j, b_j) [13]. Σ must be strictly positive definite.

G1. Group families by ordered role set; each group shares Σ.

G2. Integrate out each family's own untruncated coordinates (always the genetic rows on the public path; also an unobserved relative). Families sharing an untruncated set run together, and P, sd are formed from the marginal covariance of the kept block, so a family's draws depend only on its own bounds and seed, never on the other families in a call, round or chunk. The sweep is on the remaining truncated liabilities y; the genetic mean is $\mathbb{E}[g \mid y]$ and the reported posterior variance is $\mathrm{Var}(\mathbb{E}[g \mid y]) + \mathrm{Var}(g \mid y)$. The low-level chain `rtmvnorm_gibbs` is not collapsed.

G3. Initialise each chain at a feasible point of the rectangle. Hold pinned coordinates.

G4. Sweep free coordinates by inverse-CDF. Far-tail intervals (Φ underflows near $|z| \approx 38.5$) invert the leading-order Gaussian tail ratio on the log scale (a Rayleigh construction), an approximation.

G5. After burn-in, stream means, second moments, and batch-mean summaries. Do not store the draw array.

G6. Rerun unconverged families in rounds, each a fresh chain whose sums are pooled with earlier rounds, until every requested batch-means SE [11] is $\leq$ tol. Families that have met the tolerance are dropped from later rounds.

For strongly coupled free coordinates, step G4 additionally sweeps along whitened eigenvectors of their precision block. If $Q_{ff} = UDU^\top$, pad the columns of $UD^{-1/2}$ with zeros on pinned coordinates to form directions v_k. Each has $v_k^\top Q v_k = 1$, so its displacement has law

$$t \mid x \;\sim\; \mathcal{N}(-v_k^\top Q x, 1) \quad \text{restricted to} \quad \{t : a_j \leq x_j + t v_{kj} \leq b_j \text{ for all } j\}. \tag{10}$$

Every such move preserves the same truncated law, including conditioning on nonzero pins. The supplement activates when $\max_j Q_{jj}(Q_{ff}^{-1})_{jj} > 16$ among free coordinates; ordinary blocks retain their coordinate sweeps. It costs a precision-block eigendecomposition per family and $O(d^2)$ extra work per sweep and storage per active family, without retaining draws. The persistent chains used by the moment fitters retain their coordinate-only sweeps.

The stopping rule estimates Monte Carlo precision; it does not establish stationarity. Batch means can miss slow modes even with joint moves. Check difficult bounds with independent seeds, longer burn-in and chains, and a deterministic reference where available. The historical no-mixture benchmarks below used the coordinate sampler before this supplement.

4.2 Algorithm P (Pearson–Aitken)

If one coordinate of a multivariate normal is selected from $\mathcal{N}(m_i, v_i)$ to $(m^\star, v^\star)$, the others update by [19, 3]

$$m_j \leftarrow m_j + \frac{\Sigma_{ji}}{v_i}(m^\star - m_i), \qquad \Sigma_{jk} \leftarrow \Sigma_{jk} + \frac{\Sigma_{ji}\Sigma_{ik}}{v_i^2}(v^\star - v_i). \tag{11}$$

Mendell and Elston applied (11) sequentially to threshold traits [18].

P1. Place the target first; other rows keep the caller's order. The object API sorts roles by name, so its fold order is a property of the pedigree shape; array and kinship callers set the fold order themselves.

P2. Marginalize absent rows and condition all pins jointly (Section 5). Incompatible pins, and an interval that excludes a liability the pins determine, are refused on the standard and mixture paths alike.

P3. Fold remaining intervals last-to-first: replace each marginal by its truncated-normal moments and apply (11). One remaining interval requires one exact selection update.

P4. Apply the target's interval last. An unbounded genetic target is a no-op; full liability remains conditioned on its own interval.

P5. Read the target's updated mean and variance.

After one truncation the selected law is no longer Gaussian. Algorithm P keeps the first two moments and proceeds as if it were. Hence: exact for pins followed by at most one interval; a sequential two-moment approximation for multiple remaining intervals. On singular covariance support, low-level PA requires the remaining non-deterministic interval coordinates to be linearly independent. Dependent intervals are rejected even when feasible: moment projection does not preserve their support and can otherwise return moments for an impossible event. Singular target or unobserved rows alone are allowed.

On the historically tested no-mixture structures, average agreement was high for the posterior *mean* (Section 8). It has not been characterised for the mixture, and PA's reported conditional variance is itself an approximation. A PA standard error of zero means no Monte Carlo error.

4.3 Algorithm M (censored-control mixture)

A person unaffected only up to current follow-up is a mixture of a lifetime control and a not-yet-onset future case [14]. Algorithm P folds that person by replacing the ordinary truncated moments in P3 with the two-component moments below, retaining the original sequential treatment of pins for this mixture path. With current CIP K_i and lifetime K_{pop},

$$\pi \;=\; \frac{\Phi_{\text{below}}}{\Phi_{\text{below}} + (1 - \Phi_{\text{below}})(K_{\text{pop}} - K_i)/K_{\text{pop}}}, \tag{12}$$

where the split is the *lifetime* threshold and Φ_{below} is the current conditional mass below it. The selected moments are the corresponding two-component mean and variance. Age enters only through K_i.

The factor $(K_{\text{pop}} - K_i)/K_{\text{pop}}$ is the CIP probability that a future case has not yet onset. That is an onset-timing assumption: among people who will eventually be cases, time-to-onset

is independent of liability, and censoring is non-informative given the stratum. If higher liability advances onset, the not-yet-onset component is not the above-threshold tail assigned to it. A PA generative check at $\rho = 0.6$ finds statistically detectable but practically negligible ranking shifts and the usual downward slope shift; the pin, not the mixture, is what breaks materially (Section 8). There is still no Gibbs implementation of (12).

5 Reducing the inference problem

Let $q = h^2$. The efficient calculation begins by distinguishing information that is absent, information that is exact, and information that remains an interval. These distinctions concern the observation model, not a different estimand. Throughout this section the censoring mixture is excluded. Its sequential interpretation remains Algorithm M.

5.1 Marginalize absent observations; condition exact ones

Partition a jointly Gaussian vector into unobserved non-target coordinates U, pinned coordinates P observed at p, and retained coordinates X. The latter contain the requested target and the remaining intervals. Integrating out U simply selects the (X, P) principal covariance block; it is not conditioning U to zero. For a zero-mean prior, exact conditioning then gives

$$m_X = \Sigma_{XP}\Sigma_{PP}^{-1}p, \tag{13}$$

$$V_X = \Sigma_{XX} - \Sigma_{XP}\Sigma_{PP}^{-1}\Sigma_{PX}. \tag{14}$$

The inverse notation denotes linear solves. A singular pin block requires compatible observations in its Gaussian support, not arbitrary jitter. A zero conditional variance denotes a deterministic coordinate: its bound must contain that value. Incompatible exact observations are errors.

The covariance in (14) depends on the pedigree covariance and pin mask, not the numerical onset values. Families with the same masks can share the reduction. With no remaining interval, Gaussian conditioning is the whole calculation. With one remaining interval, the univariate selection formula is exact. With several intervals, apply PA to the centered remaining bounds, preserving their relative order. Conditioning pins first changes the old PA approximation when a pin previously followed an interval. It leaves the specified joint distribution unchanged; it does not prove that every multi-interval approximation becomes more accurate.

For family-free ADuLT, let m, v be the scalar mean and variance of the full liability conditional on its observation. Gaussian regression gives

$$\mathbb{E}[a_o \mid D_o] = qm, \qquad \mathrm{Var}(a_o \mid D_o) = q(1-q) + q^2 v. \tag{15}$$

A pinned value t gives qt and $q(1-q)$. No family covariance or numerical integration is necessary. With fixed bounds and $q > 0$, changing q only rescales this family-free phenotype. An ordinary scale-invariant linear or mixed-model association statistic therefore does not require estimating q for this use. Family-informed scores and liability-scale reporting do.

5.2 Nuclear families have at most two parental factors

Assume unrelated, noninbred parents, the additive model, and independent residual environments. Write $z = (a_m, a_f)^\mathsf{T} \sim N(0, qI_2)$. For offspring j, including the proband,

$$a_j = \tfrac{1}{2}(a_m + a_f) + M_j, \qquad M_j \overset{\text{ind}}{\sim} N(0, q/2). \tag{16}$$

The Mendelian terms are independent of z. Conditional on z, all observed full liabilities are independent normals,

$$\ell_j \mid z \sim N(b_j^\mathsf{T} z, d_j), \qquad (b_m, b_f, b_{\text{child}}) = ((1,0)^\mathsf{T}, (0,1)^\mathsf{T}, (1/2,1/2)^\mathsf{T}), \tag{17}$$

with $d_m = d_f = 1 - q$ and $d_{\text{child}} = 1 - q/2$. A missing observation contributes one. An interval contributes

$$L_j(z) = \Phi\left(\frac{u_j - b_j^\mathsf{T} z}{\sqrt{d_j}}\right) - \Phi\left(\frac{l_j - b_j^\mathsf{T} z}{\sqrt{d_j}}\right), \tag{18}$$

whereas a pin contributes the corresponding normal density at its observed value. Thus

$$p(z \mid D) \;\propto\; \phi_2(z; 0, qI_2) \prod_j L_j(z). \tag{19}$$

The number of siblings changes the number of scalar factors, not the integration dimension. If no parental observations remain, their shared mean $S = (a_m + a_f)/2 \sim N(0, q/2)$ suffices and the integral is one-dimensional. Personalized thresholds do not alter this factorization.

Pins can be absorbed before integration. For $q > 0$ and positive d_j, the conditional factor precision and mean are

$$H = (qI_2)^{-1} + \sum_{j \in P} \frac{b_j b_j^\mathsf{T}}{d_j}, \qquad m_z = H^{-1} \sum_{j \in P} \frac{b_j p_j}{d_j}. \tag{20}$$

Only unpinned interval factors remain. Integrating in a Gaussian coordinate system adapted to the posterior mode further prevents rare observations from placing most posterior mass beyond the original prior-centered nodes. The mode is found by Newton's method on the log-concave negative log posterior, with its exact gradient and Hessian and Armijo backtracking; it stops on a negligible step, a failed line search, or a decrease at floating-point resolution, and fails after 80 iterations. The tensor Gauss–Hermite grid is centred there, scaled by the inverse-Hessian Cholesky factor and importance-reweighted by prior times interval likelihoods. Products and normalizing constants are evaluated in log space.

To recover the proband, let $S = b_o^\mathsf{T} z$, $v_o = 1 - q/2$, and $r = q/2$. Let $m_o(z), v_o(z)$ be the scalar moments of ℓ_o after its own observation at fixed z; for an unobserved proband these are S, v_o. Then

$$\mu_a(z) = S + \frac{r}{v_o}\{m_o(z) - S\}, \tag{21}$$

$$V_a(z) = r - \frac{r^2}{v_o} + \frac{r^2}{v_o^2} v_o(z). \tag{22}$$

The requested mean is $\mathbb{E}_{z|D}[\mu_a(z)]$. Its posterior variance is

$$\operatorname{Var}(a_o \mid D) = \mathbb{E}_{z|D}[V_a(z)] + \operatorname{Var}_{z|D}[\mu_a(z)]. \tag{23}$$

The first term must not be dropped. Full-liability moments follow by using $m_o(z), v_o(z)$ instead. With Q nodes per factor and s observations, the two-dimensional quadrature costs $O(sQ^2)$, with $O(Q^2)$ working storage when observations are processed one at a time. Resolution starts at $Q = 16$ and doubles to a cap; each doubling records the larger change in the two moments, and a result is accepted only when both of the last two changes are within tolerance (so at least three resolutions). This is a convergence diagnostic for both moments; agreement of successive resolutions is not a certified error bound. The implementation rejects unresolved convergence rather than returning an apparently precise score.

This is a factorization of the additive nuclear-family model, not a new observation model. The initial quadrature interface excludes shared C/M environments, arbitrary/inbred pedigrees, multiple traits, and the censoring mixture. It requires $q < 1$ because a parental residual becomes singular at $q = 1$; $q = 0$ is an analytic special case. PA and Gibbs retain their broader model support.

5.3 Compute selected relationships with bounded reuse

Ancestral closure is needed to calculate exact relatedness, but its entire dense relationship matrix is not needed for a score. With parents s_i, d_i and a parent-before-child ordering, the tabular identities are

$$A_{ii} = 1 + \tfrac{1}{2}A_{s_i d_i}, \qquad A_{ij} = \tfrac{1}{2}(A_{s_i j} + A_{d_i j}) \quad (i \text{ later than } j). \tag{24}$$

An unknown parent contributes zero; distinct unrelated founders have zero relationship. Evaluate (24) only for requested pairs and their dependencies, retaining the complete ancestry graph. An explicit stack avoids recursion-depth limits. A bounded cache shares computed entries between overlapping probands; eviction changes work, not the result. This saves repeated local dense matrices but does not imply linear worst-case time: the number of required pairs can still approach the square of the population size. Transient dependency storage also depends on pedigree depth.

Preserve the current observation order when reducing a PA input. Preserve the observation counts and calendar-time masking separately from inference coordinates. Near singular parameter boundaries, covariance repair can change the effective model; the driver retains the full construction when its conservative positive-definiteness bound cannot rule out that repair.

5.4 Fit covariance parameters by pairwise likelihood

For common binary thresholds and independent families, parameter estimation need not reconstruct every latent liability. Let $\theta = (q, c^2, m^2)$, restricted to identifiable nonnegative components with positive residual variance. For a pair,

$$\rho_{ij}(\theta) = qA_{ij} + c^2 C_{ij} + m^2 M_{ij}. \tag{25}$$

Let $p_{ab}(t, \rho)$ be its bivariate-normal probability for binary outcomes $a, b \in \{0, 1\}$ at threshold t. The composite log likelihood is

$$\ell_c(\theta) = \sum_f w_f \sum_{(i,j)\in f} \log p_{y_i y_j}\{t, \rho_{ij}(\theta)\}. \tag{26}$$

The weight is one for population sampling; valid positive inverse inclusion probabilities can supply w_f under the existing IPW contract. Pairs sharing the same threshold and component design have the same four probabilities, so their weighted 2×2 tables can be accumulated once. This compresses *the composite objective*; it does not make pair counts sufficient for the full multivariate family likelihood.

At a common threshold, a stable expression for a discordant cell is

$$p_{01}(t, \rho) = \frac{1}{2\pi} \int_{\arcsin \rho}^{\pi/2} \exp\left\{ -\frac{t^2}{1 + \sin u} \right\} du. \tag{27}$$

The other cells follow from the known marginal probabilities. This avoids subtracting two nearly equal bivariate probabilities as ρ approaches one. Deterministic quadrature and analytic probability derivatives permit low-dimensional constrained optimization [1, 2].

Within-family pairs are dependent. At an identifiable interior solution, let s_f be the family score, $H = -\nabla^2 \ell_c$, and J the empirical outer-product sum of weighted family scores. The sandwich covariance is

$$\widehat{\mathrm{Var}}(\widehat{\theta}) = H^{-1} J H^{-1}. \tag{28}$$

The implementation centers family scores and applies the finite-cluster correction. It does not report the ordinary inverse Hessian as the sampling covariance. Boundary or singular-information fits do not justify an interior normal approximation and return explicitly unavailable standard errors. Family bootstrap is appropriate only under its independent-cluster sampling assumptions. Ascertainment is not repaired by changing the numerical engine; personalized/pinned thresholds and overlapping extracted register families remain outside this initial fitter's contract.

Pairwise fitting estimates the same covariance parameters with a different statistical criterion from the stochastic moment fitter. Neither matching point estimates on a few simulations nor passing probability tests establishes coverage or statistical efficiency. The new method is explicit and opt-in.

5.5 Joint genetic and environmental covariance

For $P \geq 2$ binary traits, `fit_pairwise_multi` extends the observed-pair criterion to positive-semidefinite trait covariance matrices. Write G, S, T, E for additive genetics, full sibship, couples and within-person residuals. In trait-major coordinates, equation (29) defines the model:

$$\Sigma = G \otimes A + S \otimes C + T \otimes M + E \otimes I, \qquad \operatorname{diag}(G + S + T + E) = \mathbf{1}. \tag{29}$$

Omit S and/or T when their components are not requested; E remains positive definite. Heritabilities are $\operatorname{diag}(G)$, and r_g and r_e standardise G and E, respectively. Thus r_e describes residual environment, while $S + T + E$ is total non-genetic covariance. The descriptive couple kernel does not model the inheritance consequences of assortative mating.

Each trait has one common threshold across people, though thresholds may differ between traits. Only jointly observed coordinates contribute probabilities or identifying contrasts: missing relatives cannot identify a component through latent imputation. The fitter rejects deficient observed designs, overlapping person IDs, and failed or infeasible optimisation. Independent families, population sampling or known positive family inclusion probabilities, and missingness that preserves pair distributions are required. Family-score sandwich covariance and a delta method give conditional sampling SEs; thresholds and weights are held fixed. Any covariance boundary withholds all normal SEs, and correlations with negligible component variance are undefined.

The 16 September development capsule retains repeated recovery and coverage checks for signed and null correlations, shared environment, missing observations, positive IPW, and omitted shared components. Coverage conditions on available interior SEs and does not establish boundary inference or general calibration for register data. The archived source predates the version-only change to v0.7.0. Multi-trait liability scoring still uses $G \otimes A + E \otimes I$; folding $S + T$ into its residual would change the model.

5.6 Time and memory between versions

The 9 September 2026 rerun compares clean v0.6.1 (`b516271`) with v0.6.0 (`52dec52`) using seven identical synthetic workloads. PA groups complete observation masks as byte strings, avoids identity-order copies of input bounds, and centers owned float64 reduction buffers in place. Parent-graph construction builds ordered sibling lists directly, removing temporary sets. Table 2 measures these changes with the same inference rules. Every PA mean and variance, register score and diagnostic, and both graph adjacency digests matched exactly.

Each PA batch has 200,000 families and 17 coordinates with a shared positive-definite covariance; these are numerical workloads, not simulated populations. Graphs have half founders and two children per parent pair. The register case has 300 probands, a shared paternal chain of depth 60 and degree-one observation selection. Its warm improvement is only about 3%. Separately traced call-allocation peaks fall from 179.2 to 50.2 MiB for mixed-mask PA and from 536.6 to 330.2 MiB for the million-record graph; the register peak stays at 3.1 MiB. This warmed `tracemalloc` measure excludes inputs and imports and does not capture every native/JIT allocation.

All 28 workers completed with stable source hashes. The runtime was Python 3.10.20, NumPy 2.2.6, SciPy 1.15.3 and Numba 0.67.0 on arm64 macOS with 10 logical CPUs. Numba used one thread; BLAS limits were requested at one without a runtime pool query. AC power and Low Power Mode off were checked before and after; worker-start load averages were 2.46–3.29. The run capsule retains first-call times, warm repetitions, both memory measures and provenance.

Table 2: v0.6.0 / v0.6.1 runtime and memory. Warm time is the median of five calls after the first call. Each version and workload has a fresh timing process and initially empty Numba cache. RSS includes inputs, imports, reached JIT compilation and all timing repetitions. Speedup divides the two unrounded warm medians. MiB means 2^{20} bytes.

Workload	Warm (s)	Speedup	Peak RSS (MiB)
Parent graph, 200,000 records	0.432 / 0.214	2.02×	290.5 / 255.2
Parent graph, 1,000,000 records	2.101 / 1.213	1.73×	930.1 / 690.5
PA, mixed pin masks	1.120 / 0.497	2.25×	491.9 / 337.3
PA, common pin mask	0.429 / 0.366	1.17×	495.6 / 348.0
PA, intervals only	0.393 / 0.370	1.06×	347.9 / 243.6
PA, censoring mixture	0.692 / 0.643	1.08×	447.5 / 354.7
Register scoring, 300 probands	0.087 / 0.085	1.03×	159.3 / 156.8

These repeat measurements support workload-specific computational gains, not statistical calibration or full-register scaling. These measurements describe v0.6.1, not the later joint-fitting implementation in v0.7.0.

6 Implementation

The mathematics above is small dense linear algebra on one family. The software problem is many target families, most role-defined inputs sharing a pedigree shape. Fixed-parameter scoring permits overlapping families; independence is a requirement for the parameter fitters, not for computing each conditional score.

Grouping. Families with the same role set share one Σ after column alignment. PA and Gibbs group on that key, prepare the covariance once, and apply the per-family intervals in a compiled kernel. Ordinary PA further shares pin conditioning by observation mask. Quadrature uses the role grouping for alignment but adapts its factor integral to each family's observations. It does not factor a full family covariance. The object API (`Family`/`Member`) does this grouping internally. The array API receives already-aligned bounds and is the hot path.

Three input routes. The vignette follows the same mathematics through three public interfaces (Table 3). The family route is `estimate_liability`; the specified-matrix route is `estimate_liability_from_kinship`; the population-trio driver is `estimate_liabilities`. The last is not a vectorized spelling of the first: it also constructs the pedigree and observation set.

Table 3: Public scoring interfaces and their output alignment. The register driver combines pedigree extraction, supplied population CIPs, observation masking, and pinned-onset PA; it is additive-only.

Route	Caller supplies	Returned target moments
Family roles	Family/member bounds and role labels	`pids`, `genetic`, `var`, `se`
Specified kinship	A, aligned bounds, optional C/M kernels	`(est, se, var)` tuple
Population trios	Unique IDs, parents, status/age, CIPs, use, landmarks	`probands`, `est`, `var`, observation counts

Kinship closure and the observation set. Extraction visits people within `max_degree` and then adds all recorded ancestors needed for exact kinship. Those extra people remain in the ancestry graph and are marked `closure_only`; they need not all have rows in the selected inference matrix. Their bounds are uninformative by default; the explicit opt-in is `condition_closure=True`.

A caller assembling low-level bounds must apply that mask too. The output counts distinguish the observation neighbourhood (`n_relatives`, excluding the proband), bounds actually used (`n_conditioned`), and structural additions (`n_closure_only`).

Prediction uses calendar time. The required `use` argument selects the observation model. With `use="gwas"`, own diagnosis is included. With `use="prediction"`, it is uninformative and every relative is censored at the proband's calendar landmark τ_i. For relative j with birth time b_j, recorded status d_j, and diagnosis/exit age x_j, write

$$c_{ij} = \tau_i - b_j, \qquad x_{ij} = \min(x_j, c_{ij}), \qquad d_{ij} = d_j \, \mathbf{1}\{x_j \le c_{ij}\}, \tag{30}$$

when $c_{ij} > 0$; otherwise the member is uninformative. Birth, index, and attained-age values must share one numeric unit. An earlier exit stays an earlier exit, and a post-index diagnosis becomes a control at that member's cutoff age, not the proband's attained age. Single or stratified supplied CIPs then produce pinned-onset bounds. The caller still chooses the eligible prediction cohort and horizon. The driver neither selects disease-free probands nor returns clinical risks. Other encodings, C/M, and Gibbs use the lower-level interfaces.

Algorithm G. The inner sweep is Numba-JIT'd and `prange`-parallel across families in a group. Each family's untruncated coordinates are integrated out of its sweep; genetic means are the Gaussian conditionals given the truncated draws. Batch-means summaries—running sums and sums of squares—are streamed, so Monte-Carlo-SE memory is $O(F)$ rather than $O(n_{\text{sim}}F)$. The posterior variance is the streamed second moment of the target, plus the constant residual variance of any collapsed coordinate. When seeded, each family carries its own RNG seed: the draw does not depend on which worker picked it up, nor on the chunk or batch it arrived in. Without Numba the identical code runs serially. Far-tail inverse-CDF draws switch to a leading-order log-scale Rayleigh approximation once Φ underflows.

PA. The same grouping applies. On the object path fold order is canonicalised to a sorted role set, so the result is a function of the pedigree shape, not of input row order—reproducibility, not a claim of smaller approximation error; array and kinship callers fix the order themselves (Section 4.2).

What the timings measure. Section 8 quotes a 392–518× object-path speed-up of PA over grouped Gibbs at four threads. That includes Python-side assembly. The array path, on already-aligned inputs, ran at 2.02–6.29 million families per second in the same warmed run—another 13–29× over the object path as measured at v0.4.0 on four threads. The grid has not been re-measured since; later releases sped up the object path, so the ratio on the current tree is smaller, and no committed artifact measures it. Those array rates are a hot-kernel measurement, not end-to-end register ingestion. Neither set of timings establishes throughput for the per-proband extraction and dense-kinship driver in Table 3.

Two different uncertainties. Write $\widehat{\mu}_i$ for the reported estimate of (1). Then `var` is $\mathrm{Var}(a_i \mid D_F)$, the posterior spread. It does not shrink with more Gibbs draws. `se` is the estimator's own error in $\widehat{\mu}_i$: batch-means Monte Carlo for Gibbs, identically zero for PA. A PA standard error of zero means no Monte Carlo error, not zero approximation error, and not zero posterior variance. On a seven-observation fold—the regime where PA `var` is itself an approximation—the sequential-moment variance sits within 5% of the Gibbs Monte-Carlo variance.

The vignette's total-variance check is restricted to a correctly specified population-sampled design with a common prior variance: $\mathrm{Var}(\mu_i) + \mathbb{E}[\mathrm{Var}(a_i \mid D_F)] = h^2$ in the outbred case. An uninformative history gives $\mu_i = 0$ and posterior variance h^2; it does not make their sum small. A discrepancy instead calls for checking sampling, model/threshold specification, alignment, and

approximation. Inbred targets use the prior in (7); selected cohorts and mixtures need separate checks, not the unmodified teaching assertions.

A genetic score is not a probability. For relatives-only observations in an additive-only model, the target's residual is independent of those observations. Approximating its conditional genetic distribution by its two moments (m_i, v_i) gives the cumulative crossing probability

$$q_i(t) \;\approx\; \overline{\Phi}\left(\frac{\Phi^{-1}(1-K_i(t)) - m_i}{\sqrt{v_i + r_i}}\right), \qquad r_i = \frac{1-h^2}{V_{ii}}. \tag{31}$$

Here m_i, v_i use the threshold-standardized genetic scale and $V_{ii} = 1$ for an outbred target. This is a moment approximation, not an exact Gaussian posterior. It must not reuse a GWAS score that already conditions on the outcome; C/M also invalidate the independent-residual shortcut. For a disease-free-at-a cohort in the same simple onset model, the incident quantity through age $b > a$ is instead

$$\Pr(a < T_{\text{onset}} \le b \mid T_{\text{onset}} > a, D_F) \;\approx\; \frac{q_i(b) - q_i(a)}{1 - q_i(a)}. \tag{32}$$

This formula does not condition on additional survival, residency, or entry events. A disease-specific Aalen–Johansen CIP alone does not encode the extra condition of being alive and resident at the landmark. Those processes need a matching risk model, and practical predictions need held-out calibration; the driver performs neither task.

What is not implemented. There is no Gibbs version of (12). Multi-trait liability scoring is Gibbs-only. The shipped environment kernels C and M feed back into single-trait prediction through c^2 and m^2; a directional maternal effect is not M.

7 Fitting h^2 is a different problem

Uses I and II *condition* on h^2. Use III may stop at h^2 (or r_g) itself. Bring a liability-scale estimate; if what you have is observed-scale (OLS / GREML / LDSC on 0/1), Lee et al. [16] give

$$h^2_{\text{liab}} \;=\; h^2_{\text{obs}} \cdot \frac{K(1-K)}{z^2} \cdot \frac{K(1-K)}{P(1-P)}, \qquad z = \phi(\Phi^{-1}(1-K)), \tag{33}$$

with P the sample case fraction ($P = K$ drops the second factor).

A pedigree fit of h^2 from the same family records is available, but it is not "LT-FH++ with h^2 free." The shipped h^2 and A+C+M fitters use role-defined relationship classes; the arbitrary-kinship kernels are scoring inputs, not a new fitter for overlapping extracted pedigrees. The original fitters use data augmentation: draw $\ell \mid C_F, \Sigma(h^2)$, then a damped Haseman–Elston step

$$\widehat{h^2} \;=\; \frac{\sum_{\text{pairs}} A_{ij}\, \ell_i \ell_j}{\sum_{\text{pairs}} A_{ij}^2}. \tag{34}$$

This is a stochastic-approximation fixed point, not a posterior. The number it reports as a standard error is within-chain Monte Carlo noise. Across-cohort SD is 20–30× larger (Table 12).

At the true parameter θ_0, let Z include known person-specific thresholds and observed covariates. A correctly specified observation and covariance model gives, by total expectation,

$$\mathbb{E}_{\theta_0}[\mathbb{E}_{\theta_0}[\ell_i \ell_j \mid D_F, Z, A] \mid Z, A] = h_0^2 A_{ij}. \tag{35}$$

Known heterogeneous thresholds do not invalidate this identity. They do not, by themselves, establish identification or convergence of the implemented fixed point either. The shipped fitter and its prevalence screen are validated only for a common binary threshold per trait and

independent families under the declared sampling contract. Personalised bounds and onset pins remain outside that supported scope and are refused. Exploratory failures motivated this restriction; it is not a necessity of total expectation. Ascertainment is not cured by more sweeps. The damage is severe and silent: on families selected through an affected proband with a *true* h^2 of zero, (34) returns $h^2 = 1.0$, and a realised case share of 7.6% against an asserted 5.0% (1.51×) already inflates h^2 by +0.484 (Section 8.10). The claim is therefore screened for marginal inconsistency rather than taken entirely on trust: the supplied T implies $K = 1 - \Phi(T)$, under population sampling each role's case count among the n_r families in which that role is observed is Binomial(n_r, K) (Kish effective n_r under IPW), and a cell with $n_r \geq 30$ is refused when $|z| \geq 6$ and the observed/expected case ratio lies outside [1/1.15, 1.15]. Passing this role-wise screen does not certify the joint family-pattern distribution. The same remarks apply to the multiple HE fit of (h^2, c^2, m^2), where saturation shows up as an exhausted residual rather than a component at the clamp. A family bootstrap, when clusters are independent, estimates sampling variability under the fitted design. It does not de-bias selection. The explicit `fit_pairwise` alternative in Section 5.4 has the same initial sampling and threshold scope, with a composite-likelihood criterion and family-score sandwich SEs. The following archived HE results are not tests of that new criterion.

Reweighting, where it is available. The augmentation of a *given* family with *given* statuses is already the correct conditional law; what selection corrupts is the *mix* of families. So when the inclusion probability π_f is known by design and strictly positive for every stratum, weighting each family's contribution by $w_f = 1/\pi_f$ in both the numerator and the denominator of (34) re-mixes the sample to population proportions and restores the estimator. Two conditions bind. *Positivity*: a design that samples no family from some stratum has $\pi = 0$ there, and no weight reconstructs what was never observed—proband ascertainment is exactly this case. Selection on family history is IPW-correctable only if every complete observed family pattern has a known, strictly positive inclusion probability. No ascertainment likelihood for clinic samples with zero-probability family strata is implemented; the marginal screen cannot replace one. The weighted role-marginal screen can falsify some wrong weight sets; it is not a general positivity test. *Efficiency*: at $K = 0.05$ a 50/50 case/control cohort needs weights up to 19, so the effective sample size falls well below the nominal one.

A multiplicative observed-to-liability factor such as (33) cannot substitute for this. It is a function of (K, P) alone, whereas at fixed K a true h^2 of 0.5 and of 0 both drive (34) to 1.0; no invertible constant maps both back. It also transforms an *observed-scale* estimate, and (34) is already on the liability scale. If the question is h^2 from an ascertained case/control GWAS of unrelated individuals, use LDSC or GREML and (33). If the question is h^2 from population-sampled nuclear families with one threshold, the pedigree HE is a reasonable moment estimator and is calibrated in that design (Section 8). Those are not the same analysis. The benchmark's "Lee arm" is only a diagnostic: it standardises each family role by its sample case rate, pools multi-relative HE moments, and applies the proband's sample fraction in (33). It is not an implementation or empirical evaluation of conventional unrelated-sample LDSC or GREML.

8 Evidence

Unless noted, ± is the SE of a mean across independent simulated cohorts. The numbers below are from committed benchmark artifacts (campaigns of 31 July, 1 August, 10–15 and 20 August 2026). They do not automatically validate later code. Every association panel uses independent SNPs and a lightweight marginal score statistic on non-overlapping simulated families; a real-LD, relatedness-aware mixed-model run has not been done. A locked comparison to R LTFHPlus 2.2.0 and LTFGRS 1.0.1 is in Section 8.2. Table 4 is an index of those quantities: each row is a different estimand, the middle column is its value, and the right-hand column is the design that

produced it. The rest of this section says what each value is allowed to mean.

Two ratios appear later and are not interchangeable. An *NCP ratio* is $(\overline{\chi^2}_{\text{causal}} - 1)$ over the same quantity for case/control. An *eff-N proxy* is a squared-correlation ratio to the case/control label.

Table 4: Headline quantities from the committed simulation artifacts. Each row is a different quantity. Value is that quantity: a range, a mean ± replicate SE, or (the NCP-increment row only) a paired 95% CI half-width. Setting is the design. Independent SNPs throughout; no row uses the censoring mixture (12).

Quantity	Value	Setting
Corr(PA, Gibbs) posterior means	0.9995–1.0000	27-cell classic LT-FH grid; no mixture
PA / grouped-Gibbs wall-clock	392–518×	this package, 4 threads, object path
LTFHPlus Gibbs / ltpred Gibbs wall-clock	6.79 ± 0.08×	same algorithm; 200 nuclear families
LTFGRS PA / ltpred PA wall-clock	1418 ± 30×	same algorithm; 200 nuclear families
Causal-SNP NCP ratio vs case/control	1.47 ± 0.04×	classic LT-FH; independent SNPs; 10,000 probands, 30 causal SNPs
LT-FH++ minus ADuLT NCP increment	+0.145 ± 0.016	paired 95% CI half-width; independent SNPs; known CIP, pinned onset
λ_{GC} with cohort-blind family thresholds	16.46 ± 0.40	4× prevalence trend over 30 y
Test R^2 of PGS+LT-FH / PGS / LT-FH	0.338 / 0.236 / 0.170	cross-fitted on held-out genetic liability
IPW-recovered h^2 (unweighted was 1.000)	0.468	50/50 case-control; N = 3,000; truth 0.5

8.1 The sequential approximation

On a 27-cell grid of classic LT-FH rectangles (three family structures × three h^2 × three K; five cohorts of 1,500 families), Corr(PA, Gibbs) is 0.9995–1.0000. Stress pedigrees remain ≥ 0.9991 (worst single seed 0.99916). Table 5 is the $h^2 = 0.5$, $K = 0.05$ slice. Calibration slopes of correctly specified PA sit on 1 to within simulation error (e.g. 1.006 ± 0.008 at $K = 0.05$, parents + two sibs). That is the argument for using PA as the default engine when the mixture is off. It is not a licence to treat PA and Gibbs as interchangeable once (12) is on.

At four threads on one machine the object-path PA sweep is 392–518× faster than grouped Gibbs (2,000 families, 25,000 draws). The ratio moves with thread count because Gibbs is the parallel engine. Both engines are this package. The useful intra-package statement is that the approximation is cheap enough for biobank-scale C_F relative to this Gibbs, not a hardware-independent constant. The comparison to public LTFHPlus and LTFGRS is Section 8.2.

Table 5: PA versus Gibbs, classic LT-FH bounds, $h^2 = 0.5$, $K = 0.05$, five seeds. Correlation is with simulated a_i. The eff-N proxy is not an NCP ratio.

Structure	corr. Gibbs	corr. PA	eff-N / c-c	Corr(PA, Gibbs)
parents	0.382 ± 0.009	0.382 ± 0.009	$1.36 \pm 0.03\times$	0.9999
parents + 2 sibs	0.437 ± 0.009	0.437 ± 0.009	$1.66 \pm 0.07\times$	0.9999
extended	0.420 ± 0.014	0.420 ± 0.014	$1.71 \pm 0.09\times$	0.9999

8.2 Public LTFHPlus and LTFGRS

ltpred is much more computationally efficient than both public packages when the algorithm is held fixed. The same classic LT-FH rectangles, three cohorts of 200 nuclear families, $h^2 = 0.5$, $K = 0.05$, and LTFHPlus 2.2.0's own Gibbs settings ($n_{\text{sim}} = 10^5$, burn-in 1000, tol = 0.01). The artifact records distinct LTFHPlus seeds per replicate, runtime versions, and resolved R/Numba/OMP/OpenBLAS thread settings. LTFHPlus 2.2.0 is Gibbs-only. Public Pearson–Aitken is LTFGRS 1.0.1 (`method="PA"`, `useMixture=FALSE`). Table 6 is the lock.

Wall-clock is the estimator call after in-process warmup, as a cohort total and as milliseconds per family (total/200). Peak RSS is the isolated child via `wait4` (the ldpred3 inherited-floor launcher), so it includes the interpreter.

Table 6: Same families and bounds. R packages used one `future` worker; ltpred Gibbs used four Numba threads. RMSE is against LTFHPlus Gibbs except the PA–PA sentence below. Peak RSS is the isolated process, in MiB.

Estimator	corr vs LTFHPlus	RMSE	total s	ms/fam.	peak MiB
LTFHPlus Gibbs	—	—	10.27 ± 0.06	51.33 ± 0.28	440.7 ± 0.7
LTFGRS PA	0.9999 ± 0.0000	0.0046 ± 0.0003	1.802 ± 0.011	9.01 ± 0.05	259.8 ± 0.7
ltpred Gibbs	0.9999 ± 0.0000	0.0041 ± 0.0002	1.513 ± 0.012	7.57 ± 0.06	180.1 ± 0.1
ltpred PA	0.9999 ± 0.0000	0.0046 ± 0.0003	0.00127 ± 0.00003	0.0064 ± 0.0002	178.7 ± 0.3

The Gibbs scores agree at the scale of the Monte Carlo error. ltpred PA and LTFGRS PA agree much more tightly (RMSE $= 8.7 \times 10^{-5}$): the same sequential two-moment update, not two approximations.

Table 7 is the fold-time lock (mean of the three per-replicate ratios, not the ratio of the means). Same-algorithm: LTFHPlus Gibbs is $6.79 \pm 0.08\times$ the ltpred Gibbs wall-clock; LTFGRS PA is $1418 \pm 30\times$ the ltpred PA wall-clock (R `future_lapply` versus compiled Python, not a different update). Mixing algorithms, LTFHPlus / ltpred PA is $8086 \pm 250\times$; that is not "LT-FH++, but faster." The mixture (12) is not in this comparison.

Table 7: Fold wall-clock on this machine. Mean ± SE of per-replicate ratios. 200 nuclear families; one R `future` worker; four Numba threads.

Comparison	fold	same algorithm
LTFHPlus Gibbs / ltpred Gibbs	$6.79 \pm 0.08\times$	yes
LTFGRS PA / ltpred PA	$1418 \pm 30\times$	yes
LTFHPlus Gibbs / ltpred PA	$8086 \pm 250\times$	no
LTFGRS PA / ltpred Gibbs	$1.19 \pm 0.01\times$	no

A matched one-thread rerun over 12 independently seeded replicates gives $1.751 \pm 0.012\times$ for LTFHPlus Gibbs / ltpred Gibbs and $1425 \pm 22\times$ for LTFGRS PA / ltpred PA. Thus most of the four-thread Gibbs fold is the 4:1 resource asymmetry, whereas the serial PA fold is essentially unchanged. Per-replicate load was not recorded, so these remain machine-specific ratios.

8.3 Classic LT-FH as a GWAS phenotype

Three genotype/effect/cohort replicates; 10,000 probands, 5,000 independent SNPs, 30 causal, $h^2 = 0.5$, $K = 0.05$, parents plus one sibling (Table 8). The same classic LT-FH C_F, inferred by either engine, gives an NCP ratio of $1.47 \pm 0.04\times$ over case/control, with λ_{GC} consistent with 1. A single-seed $1.53\times$ does not survive replication. The oracle (true a_i) is $8.8\times$: most of the liability is still unused, as it should be when $h^2 = 0.5$ and relatives are few.

Table 8: Independent-SNP marginal-association simulation. NCP uses $\overline{\chi^2} - 1$. Power is the fraction of causal SNPs at $p < 5 \times 10^{-8}$.

phenotype	mean causal χ^2	causal-SNP NCP ratio / c-c	power at 5e-8	λ_{GC}
case/control	39.63 ± 2.19	1.00×	41.1 ± 2.2%	1.011 ± 0.009
classic LT-FH (Gibbs)	57.58 ± 1.89	1.469 ± 0.040×	48.9 ± 4.0%	1.008 ± 0.022
classic LT-FH (PA)	57.60 ± 1.91	1.469 ± 0.039×	48.9 ± 4.0%	1.007 ± 0.021
oracle true g	337.37 ± 2.07	8.771 ± 0.573×	75.6 ± 1.1%	1.047 ± 0.015

8.4 What personalisation is worth

The integrated LT-FH++ panel uses age/sex/cohort CIP, coherent follow-up, competing death, ascertainment, and stratified null SNPs (Table 9). Onset is the CIP inverse of true ℓ, and cases are pinned at that T_i—so a case's full liability is essentially observed. Ten paired replicates. Accuracy, slope and NCP are adjusted for proband sex and birth year.

Table 9: Integrated personalised LT-FH++ independent-SNP marginal association, ten paired replicates. NCP is $(\overline{\chi^2} - 1)$ over case/control. The last column is stratified-null λ_{GC} before and after the same covariate adjustment.

phenotype	adjusted corr	adjusted slope	adjusted causal-SNP NCP ratio / c-c	stratified-null λ raw → adjusted
case/control	0.586 ± 0.008	1.123 ± 0.021	1.000 ± 0.000×	1.140 ± 0.051 → 1.019 ± 0.024
ADuLT (full personalised proband CIP, no FH)	0.600 ± 0.008	1.004 ± 0.018	1.049 ± 0.004×	1.024 ± 0.047 → 1.014 ± 0.031
LT-FH single-K	0.629 ± 0.008	1.183 ± 0.020	1.149 ± 0.007×	1.131 ± 0.046 → 0.990 ± 0.041
FH + age CIP (ablation)	0.639 ± 0.008	1.017 ± 0.016	1.189 ± 0.006×	0.997 ± 0.032 → 1.021 ± 0.042
FH + age + sex CIP (ablation)	0.639 ± 0.008	1.016 ± 0.016	1.189 ± 0.006×	0.983 ± 0.032 → 1.015 ± 0.038
FH + age + cohort CIP (ablation)	0.640 ± 0.008	1.004 ± 0.016	1.194 ± 0.006×	1.013 ± 0.034 → 1.026 ± 0.033
LT-FH++ (full age + sex + cohort CIP)	0.640 ± 0.008	1.004 ± 0.015	1.194 ± 0.006×	0.994 ± 0.035 → 1.009 ± 0.041

The paired LT-FH++ minus ADuLT increment is $+0.145 \pm 0.015$ NCP-ratio units (95% CI half-width). Age accounts for most of the step beyond classic LT-FH ($+0.040 \pm 0.008$); cohort adds a further $+0.006 \pm 0.002$; sex adds nothing to adjusted NCP (-0.0004 ± 0.0010). A prespecified sex-only panel nevertheless closes a female–minus–male mean-score-error gap of 0.051 ± 0.001. Sex-specific CIP is a calibration device in this design, not a proven power device.

Read Table 9 as: under a known CIP and a deterministic onset map, family history still adds association signal after the proband's own T_i is personalised, and most of that increment is already present once age is in the CIP. Do not read it as the increment you will see in a register, where onset is noisy and $K(t; s, b)$ is estimated. The pin-equals-liability identity is an assumption of the observation model, and the simulation honours it. A separate, no-SNP ablation on the same families (section 3 of the benchmark report) finds that replacing the pin with the interval $[T(\text{onset}), \infty)$ keeps almost all of the onset increment except at high prevalence, where the pin adds another 0.01–0.02 in the squared-correlation proxy. Most of what age of onset contributes is the lower bound, not the point mass.

8.5 Cohort confounding

Because the BLUP weights in (3) do not depend on the T_i, a *shared* threshold error is mostly a calibration shift. A *stratum-specific* error also reorders scores, and—if the stratum is correlated with a marker—induces association on that marker. Table 10 is the designed case: a secular prevalence trend plus null SNPs correlated with birth year.

At $R = 1$ there is no trend-driven inflation. At $R = 4$, cohort-blind family thresholds reach $\lambda_{GC} = 16.46 \pm 0.40$; the same family model with cohort-specific K stays at 1.007 ± 0.040. Case/control is also inflated, just less so. This is the argument for treating personalised CIP as a confounding-control device, not only as a power tweak. It is an independent-SNP illustration of the cohort component; it is not the full age/sex/cohort LT-FH++ design of Table 9.

Table 10: Stratified-null λ_{GC} under a lifetime prevalence trend of R per 30 birth years. Three replicates. Truly independent null SNPs stay near 1 for every method.

prevalence trend R per 30 y	FH + cohort-specific K	single-K FH	case/control
1	0.960 ± 0.012	0.960 ± 0.012	0.899 ± 0.034
2	0.964 ± 0.054	4.539 ± 0.253	2.008 ± 0.257
3	0.922 ± 0.057	10.275 ± 0.078	4.616 ± 0.220
4	1.007 ± 0.040	16.460 ± 0.399	7.666 ± 0.094

8.6 Calibration, and a wrong h^2

A correctly specified posterior mean is self-calibrating: the regression of true a_i on $\widehat{\mu}_i$ has slope 1. On the classic-LT-FH grid that slope sits on 1 to within simulation error (e.g. 1.006 ± 0.008 at $K = 0.05$, parents + two sibs). One replicated departure remains: at $K = 0.01$ with an extended pedigree the top-decile realised/predicted ratio is 0.946 ± 0.018, about 3 SE below 1.

In this sensitivity grid, a wrong assumed h^2 leaves the ranking almost untouched and tilts the scale. With true $h^2 = 0.5$ and $K = 0.05$, assumed h^2 from 0.2 to 0.8 sweeps the slope from 2.240 ± 0.015 down to 0.678 ± 0.006 while $\mathrm{Corr}(\widehat{\mu}, a)$ stays in 0.429–0.431 (SE ≤ 0.004). This is evidence of ranking robustness in the tested design, not a guarantee for another pedigree or observation model. Vary plausible population parameters and report both ranking and scale sensitivity.

Heavy tails, generative assortment at phenotypic $\rho \approx 0.3$, and an unmodelled sibship $c^2 = 0.15$ each move the slope by only 0.01–0.03. The ranking cost is small but not uniformly negligible: assortment costs 0.007 and sibship 0.005 in correlation, while heavy tails cost 0.016 (0.4983 ± 0.0045 against 0.4820 ± 0.0024), about 3% of the control correlation and roughly 3 SE. A factor-of-two error in prevalence moves the slope to 0.89 or 1.13. Threshold provenance is a substantial calibration hazard in this grid. These few misspecifications do not establish robustness of the Gaussian family model in general.

8.7 Complementarity with a PGS

Five replicates, 10,000 probands, 2,000 independent SNPs, 50/50 train/test, PGS trained on the train case/control GWAS (infinitesimal weights; no LD to shrink). Table 11 is scored against held-out a_i.

Table 11: PGS and classic LT-FH on a held-out test half. R^2 is against true a_i. The joint model is two-fold cross-fitted OLS on the two scores.

score	corr with held-out g	R^2
case/control label	0.343 ± 0.007	0.118 ± 0.005
PGS	0.485 ± 0.009	0.236 ± 0.009
LT-FH (PA)	0.413 ± 0.003	0.170 ± 0.003
PGS + LT-FH joint (cross-fitted OLS)	—	0.338 ± 0.009

Incremental R^2: $+0.103 \pm 0.007$ for LT-FH over PGS, $+0.168 \pm 0.018$ for PGS over LT-FH (95% CI half-widths). Observed $\mathrm{Corr}(\mathrm{PGS}, \mu) = 0.2009 \pm 0.0046$ against (9) at $p = 1$ (0.2003 ± 0.0050). Conditional independence holds by construction here. That is a verification of the identity, not a test of the failure modes. Neither the absolute R^2 nor its increment is portable without the architecture and sample assumptions: 2,000 independent SNPs are a friendlier multiple-testing regime than a genome-wide panel.

8.8 The censoring mixture

The mixture (12) is PA-only. A generative check (five replicates of 20,000 nuclear families; mid-life and old-age censoring; three onset models) finds ranking costs that are resolvable but negligible in size: four of the ten paired ΔCorr contrasts exclude zero, the largest being -0.00034 with a 95% CI half-width of 0.00011, and all ten CIs are tighter than ± 0.0002. A cost of 3×10^{-4} in correlation is statistically real and practically irrelevant; the honest statement is that the mixture does not meaningfully change ranking, not that no cost is detectable. Every Δslope CI excludes zero; the mixture always lowers the slope, by 0.008–0.027, and most strongly under heavy censoring.

The third onset model is a Gaussian copula of strength $\rho = 0.6$ between liability and the CIP quantile among lifetime cases. Higher liability then advances onset, with residual noise. Plain age truncation is no longer exact, and neither is the pin. Pinned slope falls to 0.92 under heavy censoring. The lifetime-interval slope sits at 1.13, between the crossing (1.20) and stochastic (1.03) extremes.

Case encoding still dominates calibration, but the right encoding depends on the onset model. Pin cases at their onset threshold when the crossing model is believed; use the lifetime interval when onset ages are unreliable or only partly liability-dependent. There is still no Gibbs implementation of (12), so this paragraph is not a PA–Gibbs comparison.

8.9 Moment recovery of h^2 and C

Twenty-five unascertained population cohorts, 3,000 families, parents + two sibs, $K = 0.10$, one threshold (Table 12). Bias is small relative to across-cohort SD. The fitter's Monte Carlo SE is not a sampling SE.

Table 12: Haseman–Elston data-augmentation fit of liability-scale h^2 under the design it is identified in.

true h^2	fitted mean	bias	empirical SD	reported MC SE	SD / MC SE
0.2	0.201	+0.001	0.037	0.0020	19×
0.4	0.387	-0.013	0.055	0.0022	25×
0.6	0.577	-0.023	0.063	0.0023	27×
0.8	0.786	-0.014	0.061	0.0026	24×

Joint $A + C$ on the same design (mean, emp. SD): true $(0.4, 0.2)$ recovers 0.399 (0.057), 0.190 (0.034); true $(0.6, 0)$ recovers 0.583 (0.057) and a boundary 0.013 (0.012). A point near zero is not a Type I rate. Omitting a real C is mostly a problem for interpreting h^2 (at true $c^2 = 0.3$, additive-only $\widehat{h^2} \approx 0.75$ versus ≈ 0.48 with C in the model), not for predicting a_i.

A research-only cross-trait HE, 25 cohorts, recovers $r_g = 0$ as -0.006 (SD 0.080) in the presence of phenotypic correlation 0.2, and $r_g = 0.6$ as 0.598 (SD 0.071).

8.10 What selection does to the same fit

Five replicates, 3,000 analysed families, $K = 0.05$, true $h^2 = 0.5$, nuclear pedigrees. Each replicate draws a population, applies one selection rule, and fits the selected subset; only the rule differs between rows. `random_50` keeps half the population *independently of phenotype* through the identical code path, and is the control that separates selection-on-phenotype from the machinery.

Every phenotype-selected design in Table 13 is pinned at the clamp in every replicate. The sharper cell is $h^2 = 0$: the population arm returns 0.018 and the control 0.025, while all five selected designs return 1.000—complete heritability in a population with none.

It is bias, not variance, and not an unfinished chain. The gap is +0.500 at $N = 1{,}000$, 3,000 and 10,000 while the population arm's shrinks toward zero; the same 0.9999 is reached from

Table 13: Fitted h^2 under selection, and the inverse-probability weighted refit. Truth is 0.5; † marks a design whose inclusion probability is zero for some stratum, where weighting is undefined.

selection	case share	unweighted $\widehat{h^2}$	IPW $\widehat{h^2}$
none (population)	0.051	0.515	—
random 50% (control)	0.050	0.602	—
50/50 case/control	0.495	1.000	0.468
20% case-enriched	0.198	1.000	0.521
affected proband	1.000	1.000	†
any affected member	0.291	1.000	†
affected relative only	0.000	1.000	†

$h^2_{\text{init}} \in \{0.05, 0.5, 0.95\}$, so the iteration climbs to the ceiling from below rather than failing to leave a high start.

The clamp censors the magnitude. Evaluating (34) on the selected families' own liabilities is uncensored and gives 1.69 under proband ascertainment, 3.4× the truth. That statistic is a decomposition and *not* the mechanism: at $h^2 = 0$ it sits near zero while the fitter reports 1.0. The fitter never sees those liabilities. It sees truncated draws conditioned on the selected status pattern, so under proband ascertainment every augmented proband is redrawn above T, its relatives are pulled with it, and the cross-products stay positive whatever the truth.

8.11 Register depth and prospective evidence

Both benchmarks were regenerated under the repaired observation contract. Closure-only ancestors keep uninformative bounds, and the prospective arm censors each relative at (30). Each benchmark uses five independent synthetic three-generation registers of 2,529–2,719 persons with remarriages (`benchmarks/RESULTS.md` §§20–21). The pipeline benchmark draws one consistent liability field $G \sim N(0, h^2 A)$ by Cholesky.

Extraction is exact. With the full ancestral closure, the extracted sub-pedigree's kinship equals the full-population kinship over 300 probands (max abs difference 0.0). The degree limit selects which relatives are conditioned on. It never changes the kinship among them.

Relatives beyond the named-role grammar carry information. On 300 probands per register, all relatives up to third degree give corr(est, g) 0.569 ± 0.013 against 0.498 ± 0.016 for the fixed role subset. The paired gain is $+0.0706 \pm 0.0060$, 95% CI $[+0.054, +0.087]$. Through the driver of Section 6, degree 3 gives 0.567 ± 0.029 and degree 1 gives 0.522 ± 0.031. The paired difference is $+0.0451 \pm 0.0092$ (95% CI $[+0.020, +0.071]$), and both calibration slopes are 1.05 ± 0.05. A CIP estimated from the register itself matches the oracle curve to $+0.0003 \pm 0.0003$.

The prospective result is modest. The target is diagnosis in $(40, 70]$ after index age 40, and the future-case rate is 0.081 ± 0.006. With familywise censoring the score reaches rank AUC 0.654 ± 0.021 and corr 0.167 ± 0.007. Adding relatives' post-index events gives AUC 0.697 ± 0.027, but the paired difference is unresolved at five registers on both metrics (ΔAUC $= +0.043$, 95% CI $[-0.026, +0.112]$), so no relatives'-events payoff is claimed. Leaking the proband's own future outcome is unambiguous: AUC 0.991 ± 0.002, paired Δcorr $= +0.524 \pm 0.014$. The driver scores 987 probands/s at one BLAS thread (v0.7.1).

These are simulated registers under the crossing model with a logistic CIP. The six-person vignette example checks software contracts (closure, post-index invariance), not accuracy.

9 What this does not do

The score is the additive-model projection of the family history you fed it. Household transmission, dominance, epistasis, indirect genetic effects, and generative assortment are not in A. When they are in the data, they are in μ_i.

Diagnosis and family-history reporting are part of the observation model. Parental-history Alzheimer GWAS is a documented example of survival and participation bias [26]; shallow psychiatric phenotypes can carry heritable confounding [4]. A CIP estimated from an ascertained biobank sample is the wrong $K(\cdot)$. Kaplan–Meier estimates net risk under independent censoring; competing death needs Aalen–Johansen. The mixture (12) inherits both requirements.

Related target families produce related μ_i by construction. A linear GWAS of those scores without a mixed model, or without pruning to non-overlapping families, is the analysis Zhuang et al. [27] warned about.

Three results would materially strengthen the current claims: a clean-source register benchmark with the corrected observation sets; a real-LD mixed-model GWAS with calibrated nulls among related targets; and an ascertainment likelihood recovering $h^2 \approx 0$ in the null affected-proband design. None is supplied by the current evidence. A Gibbs implementation of (12) is a separate missing feature. The LTFHPlus / LTFGRS lock (Section 8.2) covers only no-mixture classic bounds on 200 equal nuclear families. The liability-dependent onset arm is no longer among them—it is reported above at $\rho = 0.6$—but a single copula strength is not a sweep, and the pinned encoding's behaviour under weaker dependence is untested.

Software. `ltpred` is a Python implementation of the construction above (NumPy/SciPy; optional Numba). It does not run a GWAS and it does not fit a PGS. Combining μ_i with a PGS is the downstream half of use I. Developed with the support of SMARTbiomed. Heavier prototypes—sex-limited A, genetic nurture, onset-age decay of r_g, a common-factor summary of r_g, a Monte Carlo EM likelihood—are checkout-only and are not covered here. The source and this note were written together with AI. Relative to LTFHPlus 2.2.0 and LTFGRS 1.0.1 it is much more computationally efficient at the same algorithm (Section 8.2). Source: `https://github.com/bvilhjal/ltpred`. Cite the method actually used [9, 20, 21, 14] and the software.

References

[1] D. Renard, G. Molenberghs, and H. Geys. A pairwise likelihood approach to estimation in multilevel probit models. *Computational Statistics & Data Analysis*, 44:649–667, 2004. doi: 10.1016/S0167-9473(02)00263-3.

[2] M. Bravo and A. Canale. Pairwise likelihood inference for the multivariate ordered probit model. *arXiv manuscript*, 2019. https://arxiv.org/abs/1901.10186.

[3] A. C. Aitken. Note on selection from a multivariate normal population. *Proceedings of the Edinburgh Mathematical Society*, 4(2):106–110, 1935. doi:10.1017/S0013091500008063.

[4] N. Cai et al. A Perspective on heritable confounding in psychiatric genetics. *Nature Genetics*, 2026. doi:10.1038/s41588-025-02465-y.

[5] D. S. Falconer. The inheritance of liability to certain diseases, estimated from the incidence among relatives. *Annals of Human Genetics*, 29(1):51–76, 1965.

[6] L. N. Hazel. The genetic basis for constructing selection indexes. *Genetics*, 28(6):476–490, 1943.

[7] C. R. Henderson. Best linear unbiased estimation and prediction under a selection model. *Biometrics*, 31(2):423–447, 1975.

[8] C. R. Henderson. A simple method for computing the inverse of a numerator relationship matrix used in prediction of breeding values. *Biometrics*, 32(1):69–83, 1976.

[9] M. L. A. Hujoel, S. Gazal, P.-R. Loh, N. Patterson, and A. L. Price. Liability threshold modeling of case–control status and family history of disease increases association power. *Nature Genetics*, 52(5):541–547, 2020. doi:10.1038/s41588-020-0613-6.

[10] M. L. A. Hujoel, P.-R. Loh, B. M. Neale, and A. L. Price. Incorporating family history of disease improves polygenic risk scores in diverse populations. *Cell Genomics*, 2(7):100152, 2022. doi:10.1016/j.xgen.2022.100152.

[11] G. L. Jones, M. Haran, B. S. Caffo, and R. Neath. Fixed-width output analysis for Markov chain Monte Carlo. *Journal of the American Statistical Association*, 101(476):1537–1547, 2006.

[12] K. S. Kendler, H. Ohlsson, J. Sundquist, and K. Sundquist. The family genetic risk score. *JAMA Psychiatry*, 78(6):637–643, 2021. doi:10.1001/jamapsychiatry.2021.0336.

[13] J. H. Kotecha and P. M. Djurić. Gibbs sampling approach for generation of truncated multivariate Gaussian random variables. In *ICASSP*, 1999.

[14] M. Dybdahl Krebs et al. Genetic liability estimated from large-scale family data improves genetic prediction, risk score profiling, and gene mapping for major depression. *The American Journal of Human Genetics*, 111(11):2494–2509, 2024. doi:10.1016/j.ajhg.2024.09.009.

[15] M. Dybdahl Krebs et al. The relationship between genotype- and phenotype-based estimates of genetic liability to psychiatric disorders, in practice and in theory. *The American Journal of Human Genetics*, 113(1):184–201, 2026. doi:10.1016/j.ajhg.2025.11.016.

[16] S. H. Lee, N. R. Wray, M. E. Goddard, and P. M. Visscher. Estimating missing heritability for disease from genome-wide association studies. *The American Journal of Human Genetics*, 88(3):294–305, 2011.

[17] J. Z. Liu, Y. Erlich, and J. K. Pickrell. Case–control association mapping by proxy using family history of disease. *Nature Genetics*, 49(3):325–331, 2017. doi:10.1038/ng.3766.

[18] N. R. Mendell and R. C. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics*, 30(1):41–57, 1974.

[19] K. Pearson. Mathematical contributions to the theory of evolution. XI. On the influence of natural selection on the variability and correlation of organs. *Philosophical Transactions of the Royal Society A*, 200:1–66, 1903.

[20] E. M. Pedersen et al. Accounting for age of onset and family history improves power in genome-wide association studies. *The American Journal of Human Genetics*, 109(3):417–432, 2022. doi:10.1016/j.ajhg.2022.01.009.

[21] E. M. Pedersen et al. ADuLT: an efficient and robust time-to-event GWAS. *Nature Communications*, 14:5553, 2023. doi:10.1038/s41467-023-41210-z.

[22] M. R. Robinson et al. Genetic evidence of assortative mating in humans. *Nature Human Behaviour*, 1:0016, 2017.

[23] P. Sham. *Statistics in Human Genetics*. Arnold, London, 1998.

[24] H.-C. So, J. S. H. Kwan, S. S. Cherny, and P. C. Sham. Risk prediction of complex diseases from family history and known susceptibility loci, with applications for cancer screening. *The American Journal of Human Genetics*, 88(5):548–565, 2011.

[25] S. Wright. An analysis of variability in number of digits in an inbred strain of Guinea pigs. *Genetics*, 19(6):506–536, 1934.

[26] Y. Wu et al. Pervasive biases in proxy genome-wide association studies based on parental history of Alzheimer's disease. *Nature Genetics*, 56:2696–2703, 2024. doi: 10.1038/s41588-024-01963-9.

[27] Z. Zhuang et al. Relatedness, case–control imbalance and tail behaviour in family-history phenotypes. *Bioinformatics*, 2022. doi:10.1093/bioinformatics/btac459.